Post-Mortem Analysis

Denver International Airport - Automated Baggage Handling System

Author: Imad Rab

Date: November 5, 2026

Summary

Automated baggage handling systems have been implemented at numerous airports all around the world. However, the automated baggage handling system of the Denver International Airport (DIA) is considered as one of the worst projects ever realized in terms of project management within IT sector. Being a baggage transportation system designed to become the largest in the world, connecting the entire airport network via a computerized train, the system has caused delays in opening of the DIA for about 16 months due to numerous problems related to complexity of the process, change of scope, and inability to conduct proper testing. As a result, about $560 million was spent on construction of the system (with the baggage system accounting for hundreds of millions). Ultimately, the whole system has never been utilized properly, and it was eventually cancelled in 2005.

Background

During the late 1980s, Denver decided to erect the huge airport that would be the replacement for the aging Stapleton International Airport. In order to make the difference between other airports and speed up the aircraft turnaround processes, the idea of building the fully-automated airport-wide baggage system has been raised, which was to use a network of computerized trains, known as telecars, running along the destination-coded track. The construction of the system was entrusted to BAE Automated Systems, known baggage handling equipment supplier.

Timeline

When	Event
1989–1992	Denver starts construction on a new airport (DIA) to replace Stapleton, and plans its comprehensive airport automated baggage system.
1992	BAE Automated Systems is hired, but too late, to develop and install the automated system that started out just for United’s concourse but was extended to the entire airport in an accelerated schedule.
1993	Opening date delays occur due to integration failures of the system, with demonstrations showing bags being jammed, misdirected, and thrown from the telecars.
1994	There are issues during the public demonstration where there is mangled and piles of luggage. An additional manual tug and cart system is installed in the city at additional cost.

Root Causes of Failure

Unrealistic Schedule & Underestimation of Complexity
No such system had been built before on such a scale. Specialists – both those with the experience of working on Munich's significantly smaller automated system and others – told that the project is unrealizable in this schedule because this type of systems require years and months of integration and testing. None of these specialists' concerns were paid attention to; the schedule was set according to the opening date.

Scope Change in Mid-project
The municipality widened the BAE's scope of responsibilities for designing and implementing the system from one concourse to the whole airport. Besides, this happened quite late in the project and did not presuppose the revision of neither time, nor budget nor design. What is even worse is the fact that the airport building was being constructed at that time, thus limiting the baggage system to pre-established spaces and turns.

Poor Management & Communication
There had been nobody to integrate the whole project into one and be responsible for the end-to-end system; the municipality as owner of the building, BAE as contractor and the airlines all had their own different expectations.

Poor Testing
Because the schedule was limited, the system was made operational way ahead of being properly tested. That resulted in demonstrations of jamming, rerouting, and destroying of baggage as well.

Lack of Contingency Planning at the Beginning
Manual operation of the system was included later as a result of inability of automating everything on time; this required extra cost and time to implement.

Impact

• The airport had been open almost 16 months after initial scheduled opening.
• The cost overrun is reported to equal almost $1 million per day because of extra cost related to interest expenses and operations; in general, the delay of airport opening is claimed to result in the cost overrun of about half a billion dollars while the baggage system itself costed hundreds of millions;
• Outcome: the automatic system had been never implemented in full but partially and manually until being abandoned in 2005;
• The reputation: the project became a classic example of bad project management discussed in various courses worldwide.

Lessons Learned

• The schedule should be adjusted to the real complexity of the work and not the deadline; some dates cannot compress the difficult technical tasks.

• The scope should be frozen. Any significant scope changes after designing have to be accompanied by resetting of the schedule, budget, and plan – not by additional pressure.
• Integration and testing have to be done as early as possible, continuously and properly. Complex integration fails at boundaries; therefore, any time limits should not affect integration testing.
• Somebody should be responsible for the complete system in relation to all participants.
• A fallback should be planned from the very beginning as risk-mitigation measure.
• If possible, the space should be designed in a way of adaptation to the system; retrofitting is more complicated and risky task.

Recommendations (How It Could Have Been Avoided)

• Prolonged phased release: the system should be proved to work on one concourse from the very beginning
• Building a realistic schedule considering necessary amount of integration time compared with analogous project (like Munich);
• Application of formal change control that would cause the scope widening reconsideration of costs and schedule.
• Nominating the independent system integrator with authorities on the entire city, contractors and airlines.
• Funding of the contingency plan all the way through instead of its adding after some realization.

Relevance to the Smart Parking Platform Project

The example of DIA directly reinforces decisions made within our project. Our estimate includes the time devoted to testing and integration as part of our bottom-up estimating; our risk register takes into account the integration risks and the third-party risks with the backup (the predictive availability if no sensors and payment provider is available), we release a thin-slice end-to-end system first (the authentication and onboarding of operators in Sprint 1) and then extend the system to the whole airport. In brief, DIA is the counterexample to our plans.

Appendix - Sources

The analysis is based on the well-established sources about the Denver International Airport baggage handling system frequently employed in teaching project management. Examples include: Case Study – Denver International Airport Baggage Handling System (Calleam Consulting), press releases and media reports, and popular IT Project Management texts. Figures given are only rough estimates.